2016 EXAMINATIONS

Lancaster University

Part II

COMPUTING AND COMMUNICATIONS [2.5 Hours]

SCC.366 Media Coding and Processing

Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 25 marks.

Use a separate answer book for each question.

Question 1

This question addresses the JPEG encoding standard.

1.a Provide block diagrams of the encoding (2 marks) and decoding (2 marks) processes in JPEG.

[**4 marks**]

1.b Define briefly the following terms

i. Lossy and lossless data compression (1 mark).

ii. Compression ratio (1 mark).

iii. Fixed length encoding and variable length encoding (2 marks).

iv. Prefix code (1 marks).

[**5 marks**]

1.c Explain briefly

i. Why lossy data compression is commonly used in image coding:
What is the advantage? Why is the resulting data loss acceptable? (2 marks).

ii. Why the chromaticity components are sub-sampled (before the encoding process) in JPEG (3 marks).

iii. Why JPEG encoded images show block artifacts for high compression ratios (2 marks).

iv. How the compression ratios (and the resulting image quality) are controlled in JPEG (3 marks).

[**10 marks**]

1.d After the DCT step, a mapped block contains DC and AC components. Explain how these components are subsequently encoded (which symbol encoding methods are used for DC and AC components, respectively?) (2 marks). Explain your choice (4 marks).

[**6 marks**]

Total 25 marks

Question 2

This question addresses the symbol encoding processes that convert a sequence of input source symbols to a sequence of target symbols, or eventually a binary digit sequence.

2.a Consider the following source symbol generation process:
Source symbols are given as a set with 6 elements: S = {A,B,C,D,E,F}. The probability (relative frequency) of the occurrence of each symbol in S is as follows:

P(A)	P(B)	P(C)	P(D)	P(E)	P(F)
0.3	0.2	0.25	0.06	0.15	0.04

i. Design a Huffman code for this source: Show the Huffman tree and the resulting code word table (5 marks). In the designed code, the symbol `A' should be assigned with the code word `0'. Calculate the average number of bits per code word (3 marks).

ii. The Huffman code is the best prefix code in the sense that no other prefix code has a shorter average code word length than the Huffman code. Explain in this case, why we use additional symbol encoding schemes, e.g., run-length encoding or differential pulse code modulation in JPEG image coding? (2 marks)

[10 marks]

2.b Consider the following numerical symbol generation and quantization processes: Source symbols are generated as integers in [0,255], i.e., S={0, 1, ..., 255}. All symbols in S are uniformly quantized into 4 equal-sized intervals. This leads to a quantized symbol set Q(S) of size 5. Provide the quantization results for the input source sequence given below:

Input: [52, 202, 90, 26, 154, 234, 108] $\subset$ S

Quantization result: [?, ?, ?, ?, ?, ?, ?] $\subset$ Q(S)

[5 marks]

2.c Provide the results of applying run-length encoding for the three source sequences that are given below. The outputs are given as sequences of symbols in S={A,B,C,D,E } and natural numbers:

[DDDDDDEECCCCCCCCCBBBBBAAAAA]

[BBBBBBBBBBBBBBBBBBBBBBBBBB]

[EBBBBCCECCBCEADDDEACCABAADDD]

[6 marks]

Question 2 continued on next page...

Question 2 continued…

2.d Provide the results of applying differential pulse code modulation for the source sequen ces that is given below:

[0, 1, 0.5, 0.9, 0.7, 0.3, 0.5, 0.8]
[1, 1, 0.9, 0.9, 0.8, 0.1, 0.1, 0.2]

[**4 marks**]

Total 25 marks

Question 3

3.a Audio is represented in a waveform, which during digitisation is transformed from the analog to the digital domain.

i. Explain the different steps of the digitisation process. Use diagrams in your explanation if you wish. **[4 marks]**

ii. Explain the role of the Nyquist theory and low-pass filtering in bounding the sampling rate of an audio signal. **[2 marks]**

iii. The audible frequency range for humans is from 20 HZ to 20 KHz. Why do telephone encoding standards not optimise for the entire frequency range? **[2 marks]**

3.b As alternative to Pulse Code Modulation (PCM) one can use Differential Pulse Code Modulation (DPCM) for coding.

i. Describe the DPCM encoding and de-coding process (use diagrams if you wish. Describe the different components **[4 marks]**

ii. Explain the issues that can arise is low bitrates are encoded with DPCM **[2 Marks]**

3.c The MPEG-1 Layer 3 (MP3) audio coding standard exploits psycho-acoustic effects. Describe the coding steps and different masking effects exploited in each step.

[6 marks]

3.d The Natural and Rural History Unit of the National Library has an analog tape base audio archive. This consists of i.) sound recordings of animals and the natural environments, ii.) interviews of habitants of rural areas, and iii.) music recordings of major classical concerts. They want to digitise these recordings in order to better preserve them and you have been asked to propose encoding (respectively compression) formats that would optimally support this. What would you recommend for the different types of recording. Consider the trade off between information preservation and costs. **[5 marks]**

Total 25 marks

Question 4

4.a Video employs individual frames as the basic elements.

i. Explain what effects are exploited so that humans perceive them as moving pictures (i.e. continuous) **[3 marks]**

ii. MPEG-1 specifies three different frame types for normal playback. Name and explain the different frame types. **[3 marks]**

4.b MPEG encoding standards have been having a significant impact in the content production and broadcast industry.

i. What are the foremost requirements media production has on an encoding standard? **[3 Marks]**

ii. Why was it deemed necessary to impose restrictions on the implementation of MEPG-2 codecs in the context, for instance of the DVB project? **[2 Marks]**

iii. A broadcaster in considering different products for archiving their broadcast quality material. The principal decision is to either use MPEG based or DV based formats. Briefly describe the differences between both. Which one would you recommend considering quality aspects, storage requirements but also available products? **[3 Marks]**

4.c Histogram based cut-detection was one of the first methods of video analysis. Explain how it works and also discuss potential problem areas where it might not work. Could these problems be overcome; if yes how? **[5 marks]**

4.d Line tracing is a basic method in video analysis. Hough Transform can help to solve the problem of line detection.

i. Explain the basic idea underlying the Hough Transform **[3 marks]**

ii. Explain how line tracing can be used in the analysis of videos (e.g. sports videos). What are the properties that make line tracing a useful analysis method for sport? Which sports video should be good to analyse and for what kind of sports might it be less useful? **[3marks]**

Total 25 marks

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